BEN ROSS

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EDUCATION

University of Maryland, College Park, B.S. Computer Science *Expected May 2029*
Coursework: Object-Oriented Programming II (data structures), Discrete Structures, Linear Algebra, FIRE research program
Activities: UMD Loop rover team (software), App Dev Club (Technical Operations)
Honors: National Merit Finalist; SAT 1570, single sitting; 99th percentile, National Cyber League (Fall 2025)

EXPERIENCE

U.S. Army Research Laboratory, *Research Apprentice (AEOP)* *Jun 2026 – Aug 2026*

- Built a pipeline that detects GPS jamming and spoofing worldwide from the position reports aircraft and ships already broadcast (ADS-B, AIS), ingesting about 100 million a day from 75,000 aircraft. Presented at the ARL Director's poster session.
- Located emitters with kinematic checks, ADS-B integrity fields, Bayesian changepoint detection and DBSCAN clustering: about 50 jamming and 25 spoofing events a day, matching published reports for Kaliningrad and the Eastern Mediterranean.
- Built a self-hosted LLM stack so engineers could run agentic coding tools inside the lab network: a 30B model on llama.cpp on one NVIDIA A100, Open WebUI for per-user auth and logging, Caddy for TLS, and a workstation installer.

Chalgen, *Sole Developer* *May 2025 – Mar 2026*

- Built an agentic harness in which an LLM writes, runs and verifies its own forensics challenges through tool calls (disk images, PCAPs, email, EXIF, encrypted PDFs, databases), taking success from 0% with chat prompting to first-try passes.
- Benchmarked Claude Sonnet 4.5, GPT-5.1 Codex and Gemini 3 Pro on success, time and cost, and built the framework Montgomery College ran its March 2026 competition on.

Sparrows, *Lead Developer & Outreach Manager* *Jun 2023 – Aug 2026*

- Built and ran a multi-tenant school platform on Next.js, TypeScript and MongoDB serving 30+ schools and about 30,000 monthly active users, as an official partner of the Montgomery County Board of Education.
- Shipped per-school routing and admin dashboards, credential auth with bcrypt and JWT, English and Spanish localization, a PWA, and AI schedule parsing on the Gemini API, deployed on AWS Amplify with CloudWatch monitoring.

PROJECTS

UMD Loop, *rover software team (University Rover Challenge)* · ARKit, MediaPipe, Python *2026*

- Built two controller-free ways to drive the rover arm: ARKit tracking of a handheld iPhone, and 3-D hand tracking from two ordinary cameras (MediaPipe landmarks, stereo calibration, triangulation). An ArUco-marker controller is verified in simulation.

SMCS Kart, *drive-by-wire electric go-kart* · C++, Python, React · github.com/PHS-SMCS/gokart-dash *2025 – 2026*

- Pitched and built a drive-by-wire electric kart with one classmate to restructure the magnet program's robotics curriculum, then co-led the 30-student class through it in subteams. It has reached 40 mph, with working steering wheel, pedals and remote control.
- Designed the controls around one motion authority: a Teensy 4.1 owns the state machine, pedals, watchdogs and throttle, an ESP32 runs PID steer-by-wire over CAN, and the Pi dashboard is advisory only, so it can crash without affecting driving.

Herald, *open-source personal agent* · Python, SQLite · github.com/benmross/herald *2026 – present*

- Runs on a home server: ingests mail, calendar, messages, location and health data from 13 sources into one SQLite ledger and reasons over it on a schedule. Anything sent on the owner's behalf needs an approval tapped on a phone, enforced in code.

BenGPT, *LLM fine-tune*: fine-tuned Llama 3.1 8B with LoRA (LLaMA-Factory, 4- and 8-bit) on five years of personal text messages; in blind chats a close friend could not reliably tell the model from the real person. *2024 – 2025*

PUBLICATIONS

S. Su, W. Sherman, J. Terrill, Z. Kingman, E. Ferrara, **B. Ross**, et al. “Planet Falconia: Mission Possible Using Immersive Visualization to Enable Digital Twin of Remote Sensing Data.” *SIGGRAPH Asia 2026 Educator's Forum* (accepted).

LEADERSHIP

Captain & Instructor, Poolesville Computer Team: wrote and taught an 11-week ML curriculum (k-NN to Vision Transformers) and cybersecurity lessons to 40+ students twice weekly; coached the team that placed 2nd at UMD's 2026 High School Programming Contest; four of seven CyberPatriot teams reached national semifinals. *2024 – 2026*

SKILLS

Languages: Python, TypeScript/JavaScript, Java, C++, SQL, Bash
ML & data: LLM fine-tuning (LLaMA-Factory, Unsloth), llama.cpp, OpenCV, scikit-learn, DuckDB, SQLite, MongoDB
Systems: Linux, Docker, AWS, Google Cloud, Cloudflare, ROS 2, Next.js, React, Teensy, ESP32, Raspberry Pi, CAN bus